

Solution to Problem set 3: Market Failure

Market Failure

1. Specify three economic reasons for a market failure. Give one example of each reason and explain how legal intervention can solve the underlying problem.

The slides mention several reasons. For example:

- *Market power (e.g. a patent creates a monopoly, which has an incentive to reduce the quantity below the competitive one, such that there is a deadweight loss)*
- *Externalities (e.g. a positive externality leads consumers not to internalize the benefits to others, and thus the quantity purchased is too low and the price is too high. One example discussed in class was the case of Covid-19 masks).*
- *Asymmetric information – an adverse selection problem leading good products to drop out of the market. For instance, an insurance company may charge higher fees that healthy people are not willing to pay, leading them to drop out of the market.*

2. Is the following statement correct? “Monopolies are inefficient because they charge a price that is too high”. If not, explain why monopolies may be inefficient.

The statement is not fully accurate. Monopolies are inefficient not because they charge a high price per se (a price is just a transfer), but rather because the price discourage buying and then the quantity purchased is too low.

3. Explain why it may be difficult to adopt a global policy to fight off climate change, based on the problem of public goods

The cost to prevent climate change (e.g. producing less in order to pollute less) is private but the benefit is public. In other words, a stable climate is a public good – it is non-rivalrous (if I enjoy a stable climate, it doesn't derogate from the enjoyment of others) and non-excludible (I can't exclude others from enjoying it at a reasonable cost). Hence, a typical public goods problem arises, where there is under-supply of the good.

4. A bank manager wants to increase the interest rate on consumer loans. One of their advisors say that this is not a good idea because of “adverse selection”. Explain what the advisor means by this.

If the interest rate increases, the “safer” borrowers may not agree to pay so much for their loans, dropping out of the market. Only the bad borrowers remain → adverse selection.

5. James and Joe live across the street from one another. They would be happy to exchange houses, but every time one of them makes an offer – the other refuses, saying “my house is worth more”. Explain why such a market failure might occur based on a behavioral reason. *This failure might occur due to the so-called “endowment effect” – each*

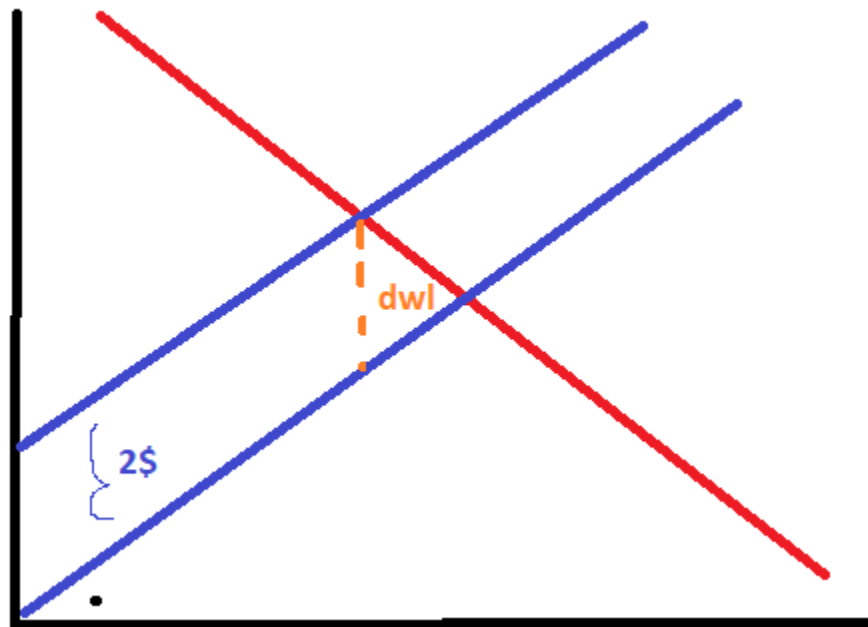
person's WTA increases once they own an item. Here, both are attached to their own house, refusing to accept the offer.

Market intervention

6. Answer the following sections (a-c):

- a. In a market with standard supply and demand curves, the government decides to impose a 2\$ tax on every produced unit. Draw a diagram of the supply and demand curves before and after the tax is imposed. Mark the area in which there is a deadweight loss, if any.

The graph is very similar to the slides (the difference is only that in the class we had a 1\$ instead of 2\$ tax. The deadweight loss is the triangle.



- b. Now suppose instead that the product is suddenly perceived as crucial for living and, as a result, the demand becomes perfectly inelastic. Explain who will bear more of the tax as a result (the supplier or the consumers) and why.

When the demand is inelastic, consumers bear more of the tax burden because they respond more sharply to a change in the price.

- c. Next, suppose instead that the supplier is a monopoly. Will this tax then exacerbate the inefficiency due to the market power or mitigate it?

The inefficiency of a monopoly is that the quantity produced is too low. A tax causes the monopoly to reduce the quantity even more, so that it exacerbates the inefficiency.

7. Which of the following statements is correct? (mark all the correct answers)
- a. Imposing a per-unit tax causes a deadweight loss. *Correct*
 - b. A per-unit tax entails a deadweight loss, but a per-unit subsidy does not. *Incorrect*
 - c. Imposing a lump sum tax does not cause a deadweight loss. *Correct*
 - d. Taxing a monopoly with a per-unit tax will cause it to produce more. *Incorrect*
 - e. A high lump sum tax might discourage firms from entering the market *Correct*
8. Suppose that the state of Hamburg imposes a fixed “Mietbremse”, where no landlord is allowed to charge more than 1500 EUR for an apartment. What will be the (possible) effects of this decision? *The excess demand combined with inability to charge higher prices might lead the seller to either sell less than before (causing people to wait in line) or try to manipulate the result by e.g. reducing the quality. If the policy is not strictly enforced, side-payments (e.g. forcing someone to buy the floor) might occur as well.*
9. Explain why maximizing the total surplus in the market for a certain product does not necessarily guarantee the maximization of social welfare. *Social welfare is the sum of utilities for everyone in society, not just buyers and sellers.*
- 10.
- a. State an economic situation with hidden information. Explain why hidden information can lead to adverse selection? What is the consequence for market efficiency? *See the example above with the bank - the bank doesn't know who is a credible borrower.*
 - b. Explain for the situation you described in (a) how signaling can improve the market outcome and why it allows the “good type” to distinguish from the “bad type”. *The good borrower may e.g. agree to deposit a security that a bad borrower cannot afford to lose. This signals that they are the “good type” and should allow the bank to overcome the problem.*